Peer Application Evaluation

ARTI 503 Parallel Architecture and Programming, Term 1 2026-2027

Subgroup being evaluated: Group 2 (Leader: Moudi Alameel)
Subgroup completing the evaluation: Group 1 (Leader: Ruaa Yasser Alghamdi)

Project summary

What is the proposed project? Summarise it in no more than 200 words.

The proposed project focuses on parallelizing a compute-intensive image-processing pipeline to evaluate the performance benefits of parallel computing. The project will compare sequential and parallel implementations of four image-processing operations: grayscale conversion, Gaussian blur, Sobel edge detection, and histogram equalization. Parallelism will be implemented both across multiple images and within individual images by dividing them into row blocks, while using overlapping border rows to maintain correct filtering results.

The project will run experiments using different image sizes, numbers of images, and numbers of workers on cloud computing resources. The results will be evaluated using execution time, speed-up, efficiency, scalability, and cost. The team will also examine synchronization and communication overheads, particularly during histogram equalization, to identify factors that limit performance gains. An optional GPU implementation of blur and Sobel operations will be tested if resources allow. The project aims to determine how effectively image-processing workloads can be parallelized and the trade-offs between performance improvement and resource cost.

Relevance

Is the topic relevant to parallel architecture and programming? Explain in approximately 200-300 words.

The proposed project is highly relevant to Parallel Architecture and Programming because it directly focuses on transforming a compute-intensive image-processing pipeline from sequential execution into parallel execution. The selected operations, including grayscale conversion, Gaussian blur, Sobel edge detection, and histogram equalization, involve processing large numbers of pixels and therefore provide suitable opportunities for parallelism.

The project demonstrates a good understanding of parallel programming concepts by considering parallelism at two levels: across independent images and within individual images by dividing them into row blocks. The use of worker processes, along with halo rows for neighborhood-based operations such as Gaussian blur and Sobel filtering, shows that the group is considering both performance and correctness when designing the parallel implementation. In addition, the proposed partial histograms and their subsequent combination introduce synchronization, allowing the group to investigate the effect of communication and synchronization overhead on performance.

The planned evaluation is also strongly aligned with the course objectives. Measuring execution time, speed-up, efficiency, scalability, and cost while varying image sizes and the number of workers provides a meaningful way to evaluate how well the parallel solution utilizes available computing resources. The comparison between sequential and parallel implementations will further demonstrate the actual benefits and limitations of parallelization.

Overall, the topic is very relevant to the course and provides a practical example of how parallel architecture and programming techniques can improve the performance of real-world computational

workloads. Including the optional GPU implementation could further strengthen the connection to parallel architectures if resources and time allow.

Clarity and writing

Is the application clear and well organised? Identify specific areas that could be improved.

The application is clear and generally well organised. The proposal follows a logical structure, beginning with the introduction and problem statement, followed by the proposed solution, budget justification, and timeline. The numbered stages in the suggested solution also make the implementation plan easy to follow. However, some information is repeated between the Problem Statement and Detailed Description, particularly the description of the image-processing pipeline. This could be shortened to make the proposal more concise. In addition, a few sentences could be revised to improve professionalism and readability. Overall, the application communicates the project effectively, but reducing repetition and improving wording consistency would make it more polished and easier to read.

Feasibility

Can the students complete the project during the course period? Is the scope appropriate?

Yes, the project appears feasible within the course period. The 13-week timeline is well structured, with clear deliverables for each stage. The scope is also appropriate because the core work focuses on sequential and parallel implementations, while the GPU implementation is optional. This makes the project manageable and realistic within the available time.

Suggested improvements

Recommend practical changes that would strengthen the application or improve its chance of support.

The project concept is sound and fits in nicely with parallel computing. However, a few changes would strengthen the implementation and results.

- The workload would increase and a more comprehensive picture of how various operations react to parallelization would be provided by adding one more operation, such as median filtering.
- It would be possible to determine how the two approaches interact and whether combining them outperforms either alone by testing a hybrid strategy that simultaneously combines both levels of parallelism—across images and within images.
- Since parallel programs frequently trade speed for higher memory consumption, adding a memory usage analysis to speed-up and efficiency would be beneficial.
- Lastly, testing on a variety of image content types, including satellite photos and medical scans, would demonstrate how well the pipeline works across real-world use cases.